Lesson plan · for the teacher

What is a sensor?

IoT · Level 1 · Lesson 1

Time
40 min

Materials
Three bowls or buckets of water: cold, room temperature and warm (not hot); A thermometer, if the school has one; 12 picture cards per group: sensors, displays and actuators; Board and chalk

Learning goals

- Explain that a sensor notices something in the world and turns it into a number.
- Name four sensors found at school, at home or in a phone.
- Sort everyday objects into sensors, displays and actuators.

Activity: Human sensors and real sensors: Card sort

1. Warm-up (5 min): Ask pupils to close their eyes for 30 seconds and notice everything they can: sounds, warmth, the chair, smells. Then list what they noticed and which body part noticed it.
2. Bowls test (10 min): A volunteer holds one hand in the cold bowl and one in the warm bowl for a minute, then puts both in the middle bowl. Repeat with two more volunteers.
3. Teach (8 min): Read the thermometer in the middle bowl, or explain it would give one number whoever held it. Write sensor, display and actuator on the board, each with a drawing and an example.
4. Card sort (12 min): Groups sort the 12 cards into three piles: sensors, displays and actuators. Walk around and ask each group to explain one tricky card.
5. Wrap-up (5 min): Each group shares one card they argued about. Exit ticket: each pupil writes one sensor, what it senses and where they have seen it.

Fill the bowls just before class; the warm water should feel comfortable, not hot. Draw 12 cards per group: thermometer, solar lamp sensor, soil probe, microphone, camera; TV screen, scale dial, scoreboard; pump, fan motor, buzzer, loudspeaker.

Discussion

- What did your ears, skin and nose notice just now?
- So why does each hand tell a different story?
- Does this card notice something, show something or do something?

Exit ticket

- One sensor + what it senses + where I saw it.

Try it in class tomorrow
Run the three-bowls test: Fill three bowls with cold, room-temperature and warm (not hot) water. A volunteer holds one hand in the cold bowl and one in the warm for a minute, then puts both in the middle bowl. Ask: "What does each hand say?" Then read a thermometer if you have one.

Worksheet · for pupils

What is a sensor?

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Name _____ Class _____ Date _____

1 **A pupil asks, "What is a sensor?" Which answer is best?**

- ◯ A part that notices something and turns it into a number or signal
- ◯ A screen that shows numbers to people
- ◯ A part that moves or switches something on

2 **Sensors are only found in expensive, high-tech gadgets.**

◯ True ◯ False

3 **Which of these is a sensor at work?**

◯ Probe in soil ◯ Water pump ◯ Wall screen ◯ Loudspeaker

4 **Match each sensor to what it notices.**

Glass thermometer	• • • •	Sound
Light sensor in a solar lamp	• • • •	How wet the soil is
Soil-moisture probe	• • • •	Daylight fading
Phone microphone	• • • •	How hot or cold it is

5 **A pupil lists parts of a smartphone. Which of them are sensors?**

- ☐ The microphone
- ☐ The camera
- ☐ The movement sensor that turns the screen
- ☐ The loudspeaker
- ☐ The vibration motor

✎ **Which sensors could your pupils find in your school or at home this week? Name two and what each one senses.**

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